

RESEARCH PAPER

Modeling Brand Visibility in Generative Engine Optimization (GEO) Using Structured Content Signals in AI Driven Search Environments

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Abstract

Search visibility is being reshaped by AI driven answer systems that synthesize responses and expose a limited set of cited sources. Traditional search engine optimization has usually been studied through ranking positions, page authority, and click oriented outcomes, while generative engine optimization requires attention to source selection, citation likelihood, and brand mention within generated answers. This paper presents an empirical modeling study of generative engine citation behavior for the purpose of developing machine learning models that predict brand visibility in AI search settings. The study uses a synthetic, controlled GEO dataset containing 43,200 query-source-engine-prompt observations, 360 queries, 3,600 candidate sources, four engine labels, and three prompt variants. Candidate sources are represented through content type, source type, authority, structure, evidence density, semantic similarity, freshness, independence, and derived retrieval features. The primary outcome is whether a candidate source is cited in a generated answer, with secondary outcomes for citation rank, brand mention, and visibility score. Statistical tests, supervised classification, ablation analysis, ranking evaluation, feature importance, and counterfactual content optimization simulations are used to assess how content signals relate to citation outcomes. The best performing classifier was logistic regression with a ROC-AUC of 0.8890, PR-AUC of 0.8995, accuracy of 0.8079, and F1 score of 0.8169 on a query grouped test set. Research reports, comparison articles, and review pages had the highest citation rates, while product and pricing pages had lower citation rates. The results suggest that content format and evidence related signals can be operationalized for GEO modeling and brand visibility analysis.

Keywords: generative engine optimization, AI search visibility, citation prediction modeling, brand visibility analysis, machine learning classification, source selection signals

1. Introduction

Search visibility has historically been organized around the idea that users enter a query, inspect a ranked list of results, and select one or more pages for further reading. In that environment, the unit of analysis is often the ranked page, and the main observable outcome is some mixture of position, impression, click, session behavior, conversion, or traffic. The technical and managerial vocabulary of search engine optimization developed around those observable surfaces. Page relevance, keyword matching, authority signals, internal linking, freshness, loading behavior, schema markup, and content quality were usually evaluated through their ability to influence ranked placement and subsequent user engagement. Although modern search engines have long used machine learning to organize results, the visible interaction still often leaves the user responsible for comparing sources and deciding what to trust.

AI driven answer systems create a different visibility problem. A user may ask a question and receive a synthesized answer that contains a small number of cited sources, a short list of supporting links, or a set of named brands embedded directly in the response. In this set-

ting, a brand may be visible even when the user never sees a conventional result page, and a source may influence the answer even when it is not the top ranked organic listing under a traditional search interface. The limited citation surface changes the practical meaning of visibility. A source that is eligible for retrieval but not selected as evidence may have little observable effect on the generated answer. A source that is selected as evidence may influence the language, recommendations, and brand comparisons that a user sees. This does not make conventional search signals irrelevant, but it means that the outcome being optimized is no longer identical to ranked placement.

The phrase generative engine optimization is used in this paper to describe the study of how content can be represented, retrieved, cited, and surfaced by AI answer systems. The term is treated here as an empirical modeling problem rather than as a prescriptive checklist. The immediate research question is not how a brand should manipulate an answer system, but how citation outcomes can be measured and predicted from observable content, query, and engine features. This framing allows a research design that separates the construction of a model-

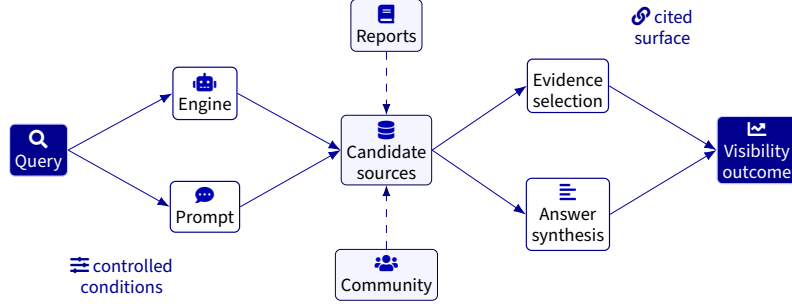


Figure 1. Generative engine visibility as a conditional pipeline from query and prompt framing to source retrieval, evidence selection, answer synthesis, and observed citation or brand visibility.

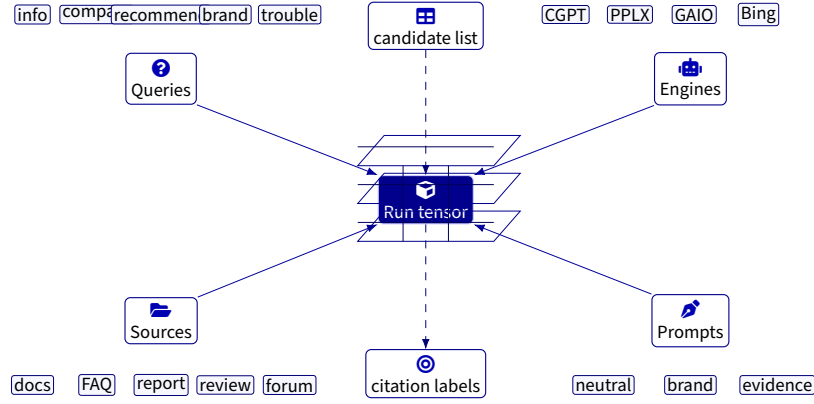


Figure 2. Experimental tensor formed by crossing queries, candidate sources, engines, and prompt variants before flattening into candidate-level modeling rows.

ing table from the later interpretation of model behavior. It also makes clear that an optimization claim must be grounded in a specific outcome variable, such as citation probability, citation rank, brand mention, or visibility score.

Generative answer systems introduce several measurement complications. The same query can be processed by different engines, and the same engine can return different sources when the prompt emphasizes evidence, comparison, or a named brand. Some systems may prefer authoritative editorial sources, while others may include community discussions or recent pages more frequently. Some content formats may be easier to cite because they contain direct comparisons, test evidence, concise tables, and topical structure (1). Other content formats may be useful for conversion or brand communication but less likely to be cited as supporting evidence in a neutral answer. A research design for GEO therefore needs to include query attributes, engine conditions, prompt variants, source attributes, and outcome labels at a level granular enough to support both statistical comparison and supervised learning (2).

This paper uses a structured GEO dataset to examine that modeling problem. The dataset was constructed as a research scaffold for GEO experiments. It includes brands, queries, candidate sources, engine runs, citation events, response level metadata, derived feature tables,

and train validation test splits. The dataset provides a controlled environment in which relationships among content type, source type, authority, structure, evidence density, semantic similarity, and citation outcomes can be studied in a reproducible way. This is useful for method development because it allows the full pipeline to be tested end to end: generating candidate features, forming labels, fitting models, validating predictions, evaluating rankings, and producing reports.

The paper is structured to keep design and results analytically separate. The Introduction defines the problem and motivates a modeling approach to generative engine visibility. The Research Design and Data section describes the data sources, unit of analysis, and variables without discussing empirical results. The Feature Engineering and Modeling Procedure section explains how observable page, query, engine, and brand attributes are transformed into modeling features. The Evaluation Strategy section explains the statistical tests, classification metrics, ranking metrics, ablation analysis, and counterfactual simulations used to evaluate the pipeline. The Results section reports the empirical outputs from the dataset and analysis pipeline. The Discussion section interprets the results in relation to GEO modeling. The Limitations section states the boundaries of the design. The Conclusion summarizes the study and closes with implications for future empirical work.

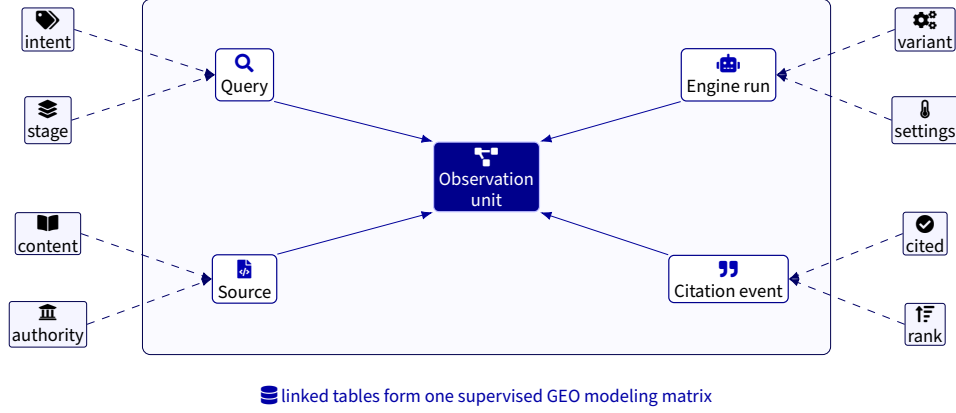


Figure 3. Linked data structure for query-source-engine-prompt observations, preserving citation as a relational outcome rather than a page-level property alone.

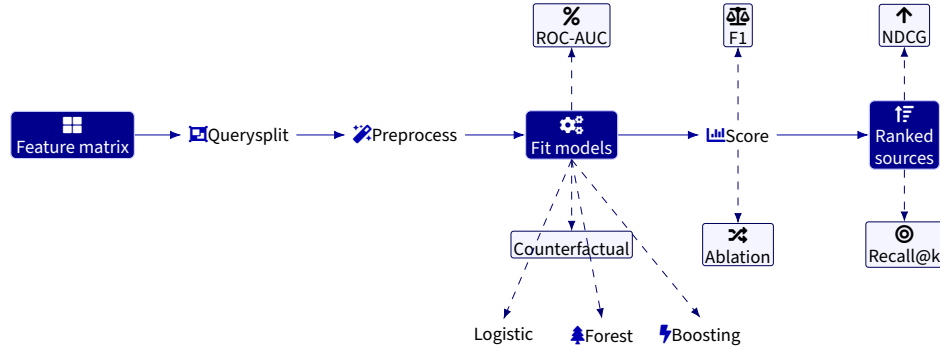


Figure 4. Supervised modeling workflow for predicting citation labels and evaluating whether cited sources are placed near the top of candidate rankings.

2. Research Design and Data

The study uses a data package designed around the query source engine prompt observation. This unit of analysis reflects the fact that citation behavior is conditional on a query, a candidate source, an engine context, and a prompt variant. A source may be cited for one query but not another, and it may be cited by one engine condition but not another. A source may also be selected under an evidence seeking prompt but not under a neutral prompt. This structure is more appropriate for generative engine research than a page level table alone, because citation is not only a property of the page. It is a relation among the source, the information need, the answer system, and the response context.

The data package contains several linked tables. The brand table represents brand level attributes such as domain, market position, authority, popularity, content maturity, schema adoption, and owned content volume. The query table represents prompts across domains, categories, intents, funnel stages, audiences, target brands, competitor brands, branded query status, query complexity, and commercial intent. The source table represents candidate pages for each query. Each source has a source type, content type, brand association, ownership flag, domain authority, word count, section count, table count, quote

count, external reference count, schema indicators, structured score, evidence density, independence score, semantic similarity, readability score, freshness score, publish age, ad density, and a composite GEO optimization score. The engine run table represents the cross product of queries, engine labels, and prompt variants, with additional meta-data such as retrieval depth, temperature, and run date. The citation event table contains the source level labels used for supervised learning.

The domains represented in the query set include SaaS, FinTech, EdTech, Cybersecurity, Travel, and Skincare. These domains were selected because they create a mix of informational, commercial, technical, and consumer oriented search settings (3). The query intents include informational, comparison, recommendation, transactional, troubleshooting, and branded prompts (4). The funnel stage variable maps these intents into broader marketing contexts, but the modeling task does not require the assumption that all user behavior follows a single linear funnel. The funnel stage is treated as a categorical feature that may help distinguish different information needs.

The engine labels are ChatGPT Search, Perplexity, Google AI Overview, and Bing Copilot. Each query is paired with three prompt variants (5). The neutral variant represents a baseline answer request. The brand ex-

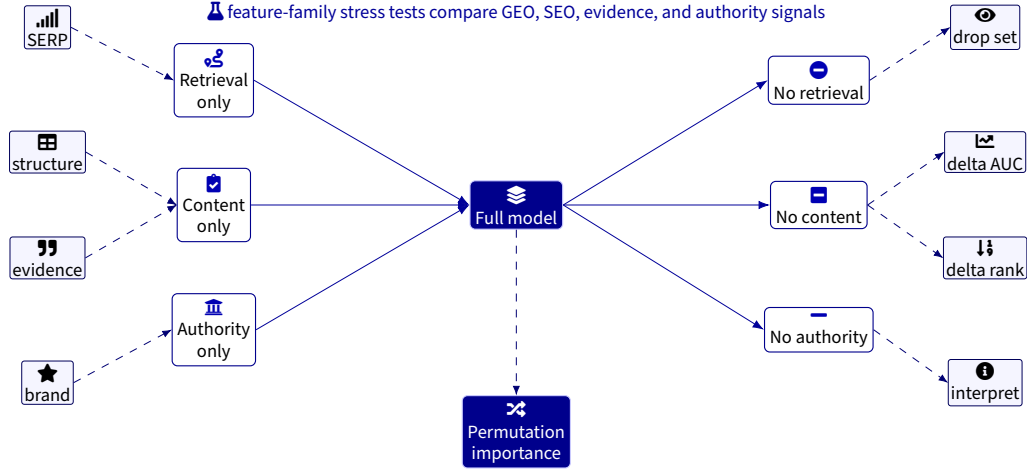


Figure 5. Ablation and permutation analysis for separating retrieval-style signals from content quality, evidence density, and authority-related features.

plicit variant names the target brand. The evidence seeking variant asks for evidence and citations. These variants create a controlled way to examine whether prompt framing interacts with source features when citation outcomes are modeled.

The source table includes content types that are common in search visibility and brand research. These include documentation, FAQ pages, comparison articles, research reports, review pages, case studies, product pages, blog posts, forum threads, news articles, and pricing pages. It also includes source types such as owned, third party, community, media, and analyst sources. These categories are intentionally broad. Their purpose is to support comparisons between content forms rather than to provide a complete taxonomy of web publishing. The distinction between content type and source type is important. A comparison article may appear on a third party site, an owned site, or a media site. A source type describes the institutional relation of the source, while a content type describes the format and informational role of the page.

The primary target variable is a binary citation label. A cited source is one that appears in the response citation list for a given run. The citation rank variable records the rank among cited sources when the source is cited. The brand mention label records whether the source brand is mentioned in the answer (6). The visibility score combines engine score, citation status, brand mention, and target brand matching into a continuous outcome (7). These labels allow the same dataset to support classification, ranking, and visibility modeling. The classification task predicts whether a candidate source will be cited (8). The ranking task evaluates whether the model can place cited sources near the top of a candidate list. The visibility task allows continuous outcome analysis, although classification is the main modeling focus in this paper.

The dataset was split using query grouped train and test logic in the modeling pipeline. Grouped splitting

is important because the same query generates multiple candidate sources, engines, and prompt variants. If observations from the same query are allowed to appear in both training and test partitions, a model may learn query specific regularities that do not generalize. Grouping by query reduces this leakage risk. The generated result files include a separate train validation test split table as well as the grouped test evaluation used by the research pipeline. The primary reported model comparison uses 32,400 training rows and 10,800 test rows.

The research design is organized around reproducibility and model evaluation. The relationships among content type, evidence density, source independence, and citation probability are represented explicitly so that the data support meaningful pipeline evaluation. The analysis demonstrates how GEO data can be structured, how target variables can be defined, and how models can be assessed across classification, ranking, and visibility outcomes. The value of the design is methodological: it provides a framework for defining variables, running tests, and reporting results that can be extended to larger or more specialized collections.

3. Feature Engineering and Modeling Procedure

The modeling table joins query, source, brand, engine run, and citation event information. This creates one row for each candidate source under a specific query, engine, and prompt condition. The feature space is divided into categorical and numeric variables. Categorical variables include engine, content type, source type, intent, domain, funnel stage, and branded query status. Numeric variables include authority related attributes, content quality attributes, retrieval related attributes, query attributes, and derived indicators for underdog brand status, ownership, and GEO optimization. This mixed feature design is appropriate because citation outcomes may depend on both discrete format categories and continuous quality

signals.

Several features are direct attributes of the source, query, brand, and engine run tables. Domain authority represents the general authority of the source domain. Brand authority and brand popularity represent brand level signals for associated sources. Word count, section count, table count, quote count, external reference count, and citation count represent visible page structure and evidentiary density. Structured score, evidence density, independence score, semantic similarity, readability score, topical coverage, ad density, and freshness score represent normalized page characteristics. Query complexity and query commerciality represent properties of the information need. These features are not intended to be the only possible GEO features. They represent a workable set that can be computed or approximated in many web data pipelines.

Derived retrieval features were added to make the v2 dataset compatible with the analysis pipeline and to represent search style covariates. A BM25 like score was derived from semantic similarity, topical coverage, and structured score. An embedding similarity feature was derived from semantic similarity and evidence density. A SERP visibility feature was derived from domain authority, backlink score, freshness, and structure. A SERP rank proxy was then derived from SERP visibility. These features allow the model to compare traditional search style signals with GEO content signals in a common pipeline.

The modeling procedure used one hot encoding for categorical variables and median imputation with scaling for numeric variables. Three classifiers were evaluated with the full feature set: logistic regression, random forest, and gradient boosting. Logistic regression provides a transparent baseline with coefficients that can be interpreted after preprocessing. Random forest provides a nonlinear tree ensemble that can capture interactions and threshold effects. Gradient boosting provides another nonlinear model that often performs strongly on structured tabular data. The comparison among these models is not intended to identify a universal best algorithm for GEO. It is intended to show how a research pipeline can evaluate several reasonable candidates under the same split and metrics.

Ablation analysis was used to assess how performance changes when feature families are restricted or removed. The full feature set includes categorical context, authority features, content quality features, retrieval features, query features, and derived GEO indicators. The retrieval only feature set uses features such as BM25 score, embedding similarity, SERP rank, SERP visibility, and semantic similarity, along with selected categorical context. The content quality only feature set uses structure, evidence, freshness, coverage, readability, word count, section count, citation count, and external references. The authority only feature set uses domain and brand authority signals, backlink score, GEO strategy bonus, and underdog status. The traditional SEO only feature set uses

authority, backlink, word count, freshness, SERP rank, and SERP visibility. The GEO content only feature set uses structure, evidence, semantic similarity, topical coverage, freshness, schema indicators, GEO strategy bonus, and content quality index. Additional ablations remove retrieval, content quality, or authority features from the full feature set.

Feature importance was assessed through permutation importance using ROC-AUC as the scoring metric. Permutation importance measures how much model performance drops when a feature is randomly permuted. It is not a causal estimate. It reflects the dependence of the fitted model on a feature given the other variables and the test sample. Logistic coefficients were also generated after preprocessing, but the paper emphasizes permutation importance because it is easier to compare across feature types. Feature correlations with the citation label and visibility score were computed separately. Correlations provide descriptive associations, while permutation importance provides model level sensitivity.

The counterfactual GEO lift simulation applies a hypothetical content optimization change to selected feature values and compares predicted probabilities before and after the change. The simulated changes include improvements to structured score, evidence density, topical coverage, freshness score, GEO strategy bonus, and ad density. The purpose is to examine how the fitted model responds when content quality signals are improved in a controlled way. The results are grouped by content type, source type, intent, engine, and underdog status to summarize where the fitted model assigns larger predicted changes.

The pipeline also produces statistical tests. A chi square test evaluates whether content type and citation status are associated. Another chi square test evaluates whether forum thread citation behavior differs across engine labels. A comparison intent test evaluates the difference between comparison articles and product pages for comparison queries. Underdog tests compare citation and visibility outcomes for underdog and non underdog brands across all rows and by content quality subset. Owned versus non owned tests compare citation rates overall and across intents. Their role is to organize patterns before and alongside model based evaluation.

4. Evaluation Strategy

The evaluation strategy uses multiple outcome views because GEO visibility is not a single number. The primary supervised learning metric is ROC-AUC, which measures the ability of a classifier to rank cited observations above non cited observations across thresholds. PR-AUC is also reported because citation can be treated as a positive event of special interest. Accuracy, precision, recall, F1 score, and Brier score are reported to characterize threshold based classification and probability calibration. Confusion matrix counts are included in the result files to

show the distribution of false positives, false negatives, true positives, and true negatives under a threshold of 0.5 (9).

The model comparison uses a grouped split by query identifier. This is essential because each query contributes multiple observations. A random row split would make it easier for the model to see similar candidates from the same query in both training and test data. The grouped split prevents direct leakage at the query level and provides a more conservative evaluation than an unrestricted row split. The reported training set contains 32,400 rows, and the test set contains 10,800 rows (10).

Ranking evaluation is included because citation prediction is often used operationally as a ranking problem. A brand or analyst may not only ask whether a source will be cited, but whether a model can identify the top candidate sources that are most likely to be cited for a given run. The ranking evaluation computes precision at k , recall at k , and normalized discounted cumulative gain at k for k values of 1, 3, 5, and 10. It compares the model ranking with a SERP visibility ranking baseline. These metrics are grouped by engine label. Precision at k reflects the proportion of top ranked model candidates that are actually cited. Recall at k reflects the proportion of all cited candidates captured within the top k . NDCG at k rewards placing cited candidates nearer the top of the ranking.

Ablation results are evaluated with the same classification metrics as the full model. The purpose is to examine whether certain feature families carry most of the predictive signal. If a restricted feature set performs nearly as well as the full set, then the omitted features may not add much incremental predictive value. If a restricted feature set performs substantially worse, then its variables are less sufficient for predicting citations.

Statistical tests are used to complement predictive evaluation. A model may perform well without providing a direct test of whether a content category differs in citation rate. Conversely, a significant content type association does not necessarily imply high predictive accuracy. The chi square tests, z tests, bootstrap confidence intervals, Welch tests, Mann Whitney tests, and effect sizes provide a separate summary of distributional patterns in the data. These tests also help identify where differences are large in practical terms and where statistical significance may occur because the sample is large.

The results include figures produced by the analysis pipeline. These figures show citation rate by content type, citation rate by engine, visibility versus semantic similarity, visibility versus evidence density, and top permutation importance features. The figures are included as external graphics in the LaTeX file. The paper does not require compilation in this workflow, but the figure paths are written so that a later LaTeX build from the project root can include the images if the usual graphics package is available in the compilation environment.

The evaluation strategy emphasizes both predictive

performance and interpretability. The main question is whether the modeling workflow can estimate citation outcomes from structured source, query, engine, and prompt variables. The reported metrics are used to compare model families, feature groups, ranking behavior, and predicted visibility changes under the same evaluation design.

5. Results

The validation output shows that the modeling table contains 43,200 rows, 360 queries, 3,600 candidate sources, and four engine labels. The overall citation rate is 0.5238, which corresponds to 52.38% of query source engine prompt observations being labeled as cited. The missing visibility rate is 0.0000, indicating that the continuous visibility score is available for all observations. The average number of candidates per query engine combination is 10.0000. The duplicate query source engine row count is 28,800 because the table includes three prompt variants for each query source engine combination. This duplication is expected from the design and is not treated as a data quality defect.

Citation rates varied strongly by content type. Research reports had the highest citation rate at 0.9053, followed by comparison articles at 0.8736 and review pages at 0.8000. Documentation pages had a citation rate of 0.5640, and FAQ pages had a citation rate of 0.5119. Lower citation rates appeared for news articles at 0.3567, forum threads at 0.3516, case studies at 0.3267, blog posts at 0.2216, product pages at 0.1393, and pricing pages at 0.0741 (11). Average visibility followed a similar ordering for the highest cited formats. Research reports had an average visibility score of 64.2058, comparison articles had 61.2982, and review pages had 55.6993. Product pages and pricing pages had lower average visibility scores of 18.1225 and 12.7727.

Engine level citation rates were close across the four engine labels. Bing Copilot had a citation rate of 0.5248, Perplexity had 0.5243, ChatGPT Search had 0.5241, and Google AI Overview had 0.5219 (12). Average visibility differed more than the citation rate. Perplexity had an average visibility score of 41.4693, ChatGPT Search had 39.7987, Google AI Overview had 39.5615, and Bing Copilot had 38.5741. Brand mention rates were also close across engines, ranging from 0.2270 to 0.2334. The domain authority average is identical across engine groups because each engine condition observes the same candidate source pool.

Source type summaries show smaller but still visible differences. Third party sources had the highest citation rate at 0.5591, followed by community sources at 0.5521 and analyst sources at 0.5489. Media sources had a citation rate of 0.4955, while owned sources had a citation rate of 0.4927. Owned sources had the highest brand mention rate at 0.3668, while third party, community, analyst, and media sources had lower brand mention rates

Table 1. Validation summary for the GEO modeling table

Check	Value
Rows	43,200
Queries	360
Candidate sources	3,600
Engines	4
Overall citation rate	0.5238
Missing visibility rate	0.0000
Duplicate query source engine rows	28,800
Average candidates per query engine	10.0000

Table 2. Citation outcomes by content type

Content type	Citation rate	Avg. visibility	Brand mention rate	Rows
Research report	0.9053	64.2058	0.2998	3,696
Comparison article	0.8736	61.2982	0.3053	6,984
Review page	0.8000	55.6993	0.2803	4,956
Documentation	0.5640	41.9089	0.2945	5,232
FAQ	0.5119	39.6567	0.2535	3,768
News article	0.3567	28.2101	0.1376	1,752
Forum thread	0.3516	27.1491	0.1477	2,796
Case study	0.3267	28.8125	0.2106	3,456
Blog post	0.2216	20.5150	0.1132	3,516
Product page	0.1393	18.1225	0.1618	4,560
Pricing page	0.0741	12.7727	0.1204	2,484

between 0.1367 and 0.1545. This pattern separates citation likelihood from brand mention likelihood. Owned sources are not the highest cited source type in the table, but they are more likely to produce brand mention when they are part of the answer context.

The content type chi square test indicated an association between content type and citation status. The chi square statistic was 14,206.2828 with 10 degrees of freedom, and the reported p value was 0.0000. The effect size was Cramer’s V of 0.5735 (13). This indicates a large distributional difference across content categories. The engine forum citation behavior test also indicated a statistically detectable association among engine labels for forum thread rows. The chi square statistic was 28.1690 with 3 degrees of freedom, the p value was 0.00000335, and Cramer’s V was 0.1004. This effect size is much smaller than the content type association.

For comparison intent queries, comparison articles and product pages differed sharply. The citation rate for comparison articles was 0.9173, while the citation rate for product pages was 0.1165. The rate difference was 0.8009, and the z statistic was 40.8996 with a p value reported as 0.0000. The mean visibility difference was 49.7524, with a bootstrap 95% interval from 48.4054 to 51.1053. This result is consistent with the role of comparison articles as a format that directly matches comparison oriented user intent.

Underdog brand tests showed mixed patterns depend-

ing on the subset and outcome. Across all rows, underdog associated observations had a citation rate of 0.5057, while non underdog observations had a citation rate of 0.5275. The citation rate difference was -0.0217 with a p value of 0.000686. However, underdog observations had higher average visibility, 41.1731 compared with 39.5806, with a mean difference of 1.5925 and a bootstrap 95% interval from 0.8961 to 2.3536. Within the high content quality subset, underdog observations had a citation rate of 0.7847 compared with 0.7725 for non underdog observations. That citation difference was 0.0122 and was not statistically significant at the 0.05 level, with p equal to 0.1135. The high quality underdog visibility difference was 4.3137, with a bootstrap 95% interval from 3.3772 to 5.2081. Within the low content quality subset, underdog observations had a citation rate of 0.2487 compared with 0.2783 for non underdog observations, with a p value of 0.000197. The visibility difference in the low quality subset was 0.5359, with a confidence interval crossing zero.

Owned versus non owned source tests showed that owned sources had lower citation rates overall but higher brand mention rates in the descriptive table. The overall owned citation rate was 0.4927, while the non owned citation rate was 0.5429. The difference was -0.0502, and the p value was reported as 0.0000. By intent, the owned source citation deficit was largest for branded queries, where owned sources had a citation rate of 0.4648 and non owned sources had 0.5600. The difference was -0.0952

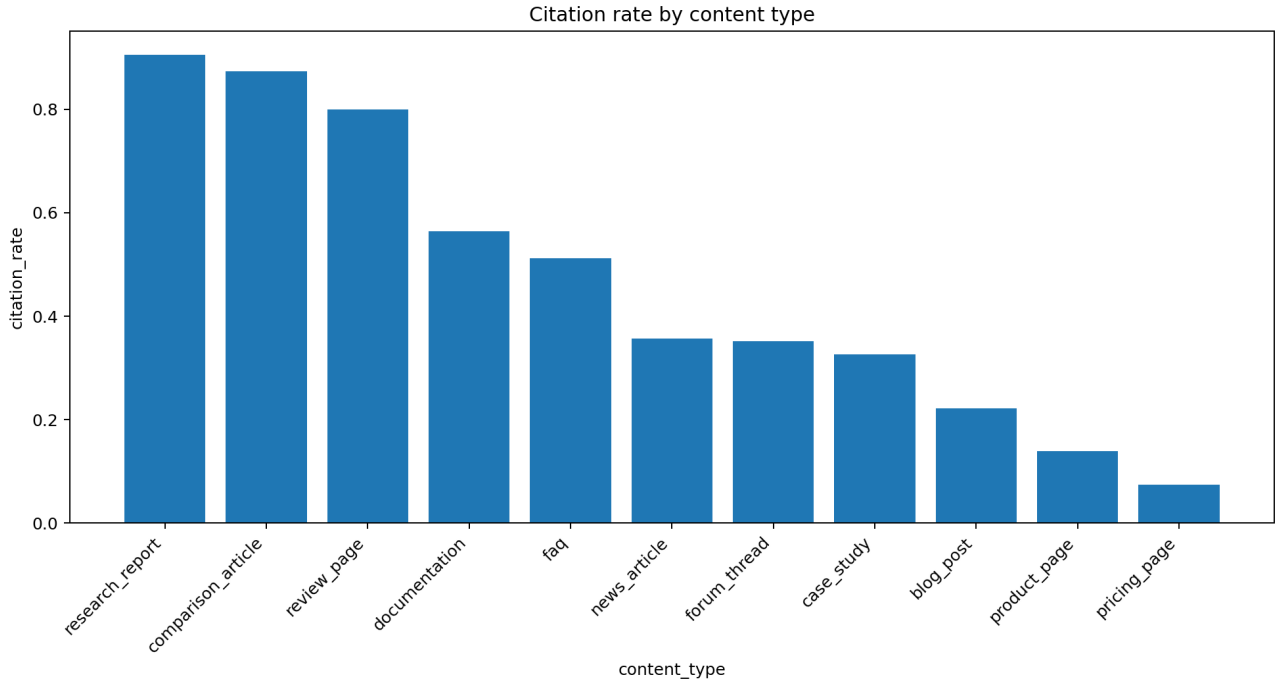


Figure 6. Citation rate by content type

Table 3. Citation outcomes by engine label

Engine	Citation rate	Avg. visibility	Brand mention rate	Rows
Bing Copilot	0.5248	38.5741	0.2322	10,800
Perplexity	0.5243	41.4693	0.2334	10,800
ChatGPT Search	0.5241	39.7987	0.2279	10,800
Google AI Overview	0.5219	39.5615	0.2270	10,800

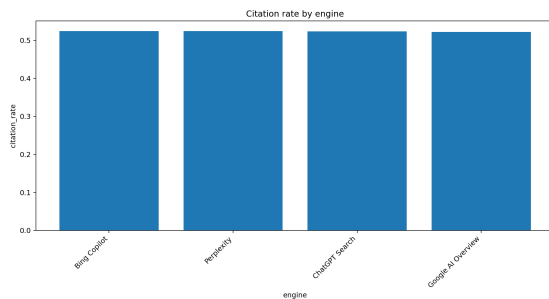


Figure 7. Citation rate by engine label

with a p value of 0.0000000000247. For recommendation intent, owned sources had 0.4701 and non owned sources had 0.5512, with a difference of -0.0811. For informational intent, the difference was smaller and not statistically significant at the 0.05 level, with owned sources at 0.5159, non owned sources at 0.5301, and p equal to 0.1572.

The model comparison results show that logistic regression had the highest ROC-AUC among the three evaluated classifiers. Logistic regression achieved ROC-AUC

0.8890, PR-AUC 0.8995, accuracy 0.8079, precision 0.8181, recall 0.8157, F1 score 0.8169, and Brier score 0.1349. Gradient boosting achieved ROC-AUC 0.8841, PR-AUC 0.8948, accuracy 0.7962, precision 0.7960, recall 0.8231, F1 score 0.8093, and Brier score 0.1386. Random forest achieved ROC-AUC 0.8800, PR-AUC 0.8906, accuracy 0.7931, precision 0.8011, recall 0.8067, F1 score 0.8039, and Brier score 0.1409. All three models were trained on 32,400 rows and tested on 10,800 rows using grouped splitting by query identifier.

The classification report for the best model gives a class 0 precision of 0.7966, recall of 0.7992, and F1 score of 0.7979 on 5,125 non cited observations. For class 1, the model had precision 0.8181, recall 0.8157, and F1 score 0.8169 on 5,675 cited observations. The macro average F1 score was 0.8074, and the weighted average F1 score was 0.8079. The confusion matrix values reported in the model comparison table were 4,096 true negatives, 1,029 false positives, 1,046 false negatives, and 4,629 true positives for logistic regression.

Ablation results show that the full feature set performed best by ROC-AUC, but several reduced feature configurations performed very close to the full model. Re-

Table 4. Citation outcomes by source type

Source type	Citation rate	Avg. visibility	Brand mention rate	Rows
Third party	0.5591	40.0755	0.1545	11,256
Community	0.5521	39.7453	0.1407	6,348
Analyst	0.5489	39.5345	0.1414	3,600
Media	0.4955	36.6915	0.1367	5,508
Owned	0.4927	40.8626	0.3668	16,488

Table 5. Underdog brand outcome comparisons

Subset	Underdog citation	Non underdog citation	Visibility difference	Citation p value
All rows	0.5057	0.5275	1.5925	0.000686
High content quality	0.7847	0.7725	4.3137	0.113451
Low content quality	0.2487	0.2783	0.5359	0.000197

moving retrieval features produced ROC-AUC 0.8890 with an AUC drop of 0.000005. Removing authority and brand features produced ROC-AUC 0.8889 with an AUC drop of 0.000051. Removing content quality features produced ROC-AUC 0.8887 with an AUC drop of 0.000323. The GEO content only feature set achieved ROC-AUC 0.8869, PR-AUC 0.8960, and F1 score 0.8165, with an AUC drop of 0.0021. The content quality only feature set achieved ROC-AUC 0.8841. Authority only achieved ROC-AUC 0.8557, retrieval only achieved 0.8539, and traditional SEO only achieved 0.8473.

Permutation importance identified content type as the dominant feature in the fitted logistic regression model. Permuting content type produced a mean ROC-AUC drop of 0.0999 with a standard deviation of 0.0099. The next highest features had smaller importance values. Independence score had a mean drop of 0.0072, SERP visibility had 0.0043, evidence density had 0.0042, backlink score had 0.0040, ad density had 0.0032, domain authority had 0.0030, content quality index had 0.0030, GEO strategy bonus had 0.0027, and topical coverage had 0.0024 (14). Structured score and query commerciality had smaller mean drops in the displayed top feature set.

Feature correlations show that the latent engine score had the strongest correlation with citation and visibility outcomes. Engine score had a Pearson correlation of 0.7237 with the cited label and a Spearman correlation of 0.8905 with visibility score. Topical coverage had a Pearson correlation of 0.5637 with cited and a Spearman correlation of 0.6754 with visibility. Content quality index had 0.5583 and 0.6708. GEO strategy bonus had 0.5360 and 0.6353. Evidence density had 0.4639 and 0.5412. Embedding similarity and BM25 score had citation correlations of 0.4079 and 0.4028. Structured score, independence score, and semantic similarity had lower but still positive correlations in the displayed results.

Ranking results show that the fitted model placed cited sources near the top of candidate lists. For ChatGPT Search, model precision at 1 was 1.0000, precision at 3 was 0.9815, precision at 5 was 0.9644, and precision at

10 was 0.9389. The corresponding SERP precision values were 0.7111, 0.7037, 0.6733, and 0.6322. For Perplexity, model precision at 1 was 0.9778, at 3 was 0.9815, at 5 was 0.9644, and at 10 was 0.9278. For Google AI Overview, model precision at 1 was 0.9778, at 3 was 0.9741, at 5 was 0.9600, and at 10 was 0.9300. For Bing Copilot, model precision at 1 and 3 were both 0.9889, at 5 was 0.9578, and at 10 was 0.9267. In all four engine groups and all displayed k values, model precision exceeded the SERP precision baseline.

Counterfactual simulations estimated larger predicted lifts for content types that started from lower baseline probabilities and could benefit from improvements to structured and evidence related signals. Forum threads had a base probability of 0.3227 and an optimized probability of 0.5152, producing an average predicted lift of 0.1925. Case studies had a base probability of 0.2669 and an optimized probability of 0.4572, producing a lift of 0.1903. FAQ pages had a base probability of 0.4685 and an optimized probability of 0.6494, producing a lift of 0.1810. News articles had a lift of 0.1783, documentation had 0.1754, and blog posts had 0.1730. By intent, troubleshooting had an average predicted lift of 0.1462 and informational had 0.1425. By source type, analyst sources had a lift of 0.1382 and media sources had 0.1342.

Taken together, the results show a coherent empirical pattern (15). Content type is the most visible differentiator in citation outcomes. Evidence oriented and comparison oriented formats are cited more often than product and pricing pages. The best model has strong discrimination under a grouped test split, and the reduced feature analyses show that several GEO content feature families retain much of the predictive performance. Source ownership is associated with higher brand mention but lower citation rate in the aggregate. The counterfactual simulation assigns larger predicted gains to content categories with lower starting probabilities and improvable structure or evidence features. These results are consistent with the study’s goal of creating a reproducible GEO modeling workflow.

Table 6. Model comparison on the grouped test set

Model	ROC-AUC	PR-AUC	Accuracy	Precision	Recall	F1
Logistic regression	0.8890	0.8995	0.8079	0.8181	0.8157	0.8169
Gradient boosting	0.8841	0.8948	0.7962	0.7960	0.8231	0.8093
Random forest	0.8800	0.8906	0.7931	0.8011	0.8067	0.8039

Table 7. Ablation results for selected feature sets

Feature set	ROC-AUC	PR-AUC	F1	AUC drop
Full	0.8890	0.8995	0.8169	0.0000
No retrieval features	0.8890	0.8995	0.8169	0.000005
No authority brand features	0.8889	0.8993	0.8188	0.000051
No content quality features	0.8887	0.8992	0.8163	0.000323
GEO content only	0.8869	0.8960	0.8165	0.0021
Content quality only	0.8841	0.8924	0.8130	0.0049
Authority only	0.8557	0.8688	0.7841	0.0333
Retrieval only	0.8539	0.8654	0.7824	0.0351
Traditional SEO only	0.8473	0.8525	0.7869	0.0417

6. Discussion

The results indicate that citation prediction can be framed as a structured tabular learning problem when query, source, engine, prompt, and brand attributes are joined at the observation level. The strongest empirical pattern is the role of content type. Research reports, comparison articles, and review pages were cited at much higher rates than product pages and pricing pages. AI generated answers often need supporting evidence, comparative context, and concise claims that can be attributed. Content formats that already organize information in those ways may be easier to select as support within a generated response.

The difference between owned source citation and owned source brand mention is an important modeling distinction. In the results, owned sources were not the highest cited source type. Third party, community, and analyst sources had higher citation rates. However, owned sources had higher brand mention rates. This implies that a brand visibility model should not treat citation and brand mention as the same outcome. A source can be cited without strongly advancing brand visibility, and a brand can be mentioned through content that is not necessarily selected as the most independent evidence. For applied GEO measurement, this suggests that a dashboard or research design should track several outcomes simultaneously rather than using citation count alone.

The model comparison results are notable because logistic regression slightly outperformed the nonlinear models. This does not mean logistic regression is generally superior for GEO. It means that the observed relationships in this dataset are sufficiently captured by a linear model after categorical encoding and feature engineering. The result is useful for research design because a transparent model can provide strong baseline performance.

Before using more complex models, researchers can evaluate whether simpler models already capture most of the signal. A simpler model may also make ablation, coefficient inspection, and reporting easier, especially in early stage studies.

The ablation results suggest that content and GEO related features can carry much of the predictive signal. The GEO content only feature set performed close to the full model, and content quality only also performed strongly. Traditional SEO only, retrieval only, and authority only feature sets performed lower than the full model. This pattern supports a modeling distinction between traditional visibility features and citation readiness features. Traditional authority and retrieval proxies remain useful, but they do not fully account for citation outcomes in the reported models. Structure, evidence density, semantic fit, topical coverage, freshness, schema indicators, and content type appear central to predicting which pages are selected as sources.

The ranking metrics are relevant for practical use. A citation classifier becomes more useful when it can rank candidate sources for a query or response context. The model achieved high precision at top k across all engine labels, and its ranking precision exceeded the SERP visibility baseline. This suggests that a GEO oriented model can be evaluated as a reranking layer. Instead of asking whether a page has good general search visibility, the model asks whether the page is likely to be cited in a generative answer for a specific information need. This distinction is important because a page can have general authority but still be a poor source for a particular generated answer.

The counterfactual simulation shows how a fitted model can be used to generate hypotheses for content improvement. Forum threads, case studies, FAQ pages, news ar-

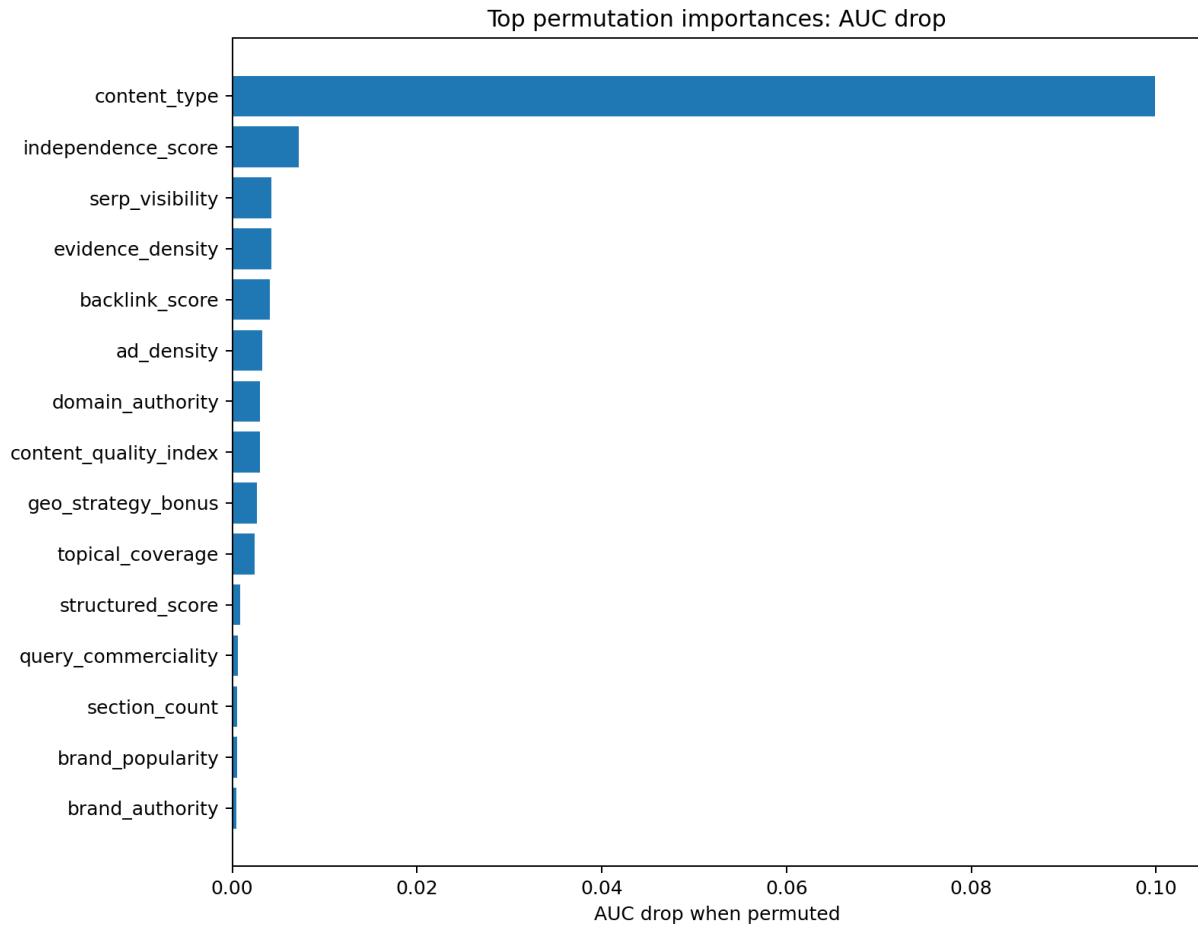


Figure 8. Top permutation importance features for citation prediction

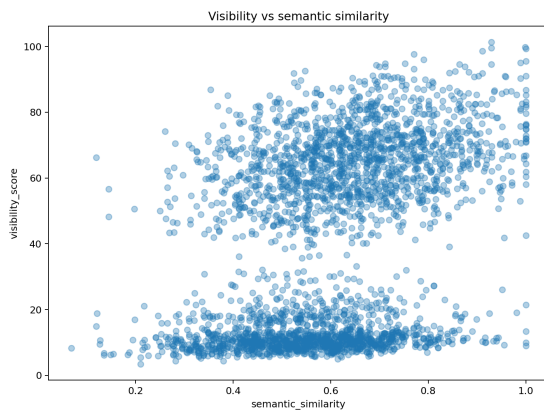


Figure 9. Visibility score and semantic similarity

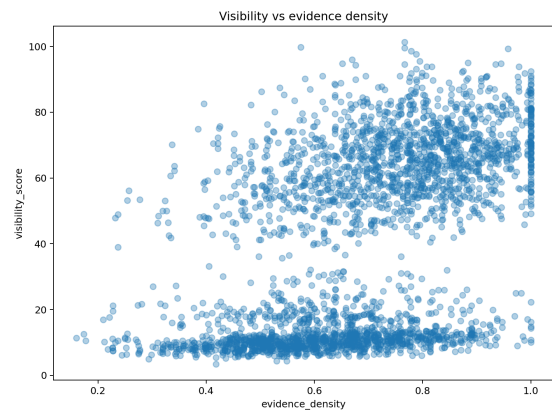


Figure 10. Visibility score and evidence density

ticles, documentation, and blog posts had relatively large predicted lifts under the simulated optimization. These categories may contain information that can become more citation ready when structure, evidence, topical coverage, freshness, and ad density are improved. The simulation is most useful as a prioritization tool for deciding which content categories and features warrant further testing.

The underdog brand analysis also requires a careful interpretation. Across all rows, underdog associated observations had a lower citation rate but higher average visibility. In the high content quality subset, underdog observations had slightly higher citation rates and higher visibility, though the citation rate difference was not statistically significant. This pattern suggests that brand po-

Table 8. Selected ranking metrics by engine

Engine	k	Model precision	Model recall	Model NDCG
Bing Copilot	1	0.9889	0.0637	0.9889
Bing Copilot	5	0.9578	0.3081	0.9673
ChatGPT Search	1	1.0000	0.0644	1.0000
ChatGPT Search	5	0.9644	0.3096	0.9732
Google AI Overview	1	0.9778	0.0621	0.9778
Google AI Overview	5	0.9600	0.3044	0.9650
Perplexity	1	0.9778	0.0628	0.9778
Perplexity	5	0.9644	0.3098	0.9690

Table 9. Counterfactual GEO lift simulation

Group type	Group value	Base probability	Optimized probability	Avg. lift
Content type	Forum thread	0.3227	0.5152	0.1925
Content type	Case study	0.2669	0.4572	0.1903
Content type	FAQ	0.4685	0.6494	0.1810
Content type	News article	0.3710	0.5494	0.1783
Content type	Documentation	0.5513	0.7267	0.1754
Content type	Blog post	0.2069	0.3799	0.1730
Intent	Troubleshooting	0.5030	0.6492	0.1462
Intent	Informational	0.4884	0.6309	0.1425
Source type	Analyst	0.5130	0.6512	0.1382
Source type	Media	0.4736	0.6078	0.1342

sition alone is not sufficient to explain visibility. Content quality and context interact with brand characteristics. For smaller or less authoritative brands, structured and evidence rich content may improve the visibility score, but it does not erase the broader relationship between authority and citation probability across all observations (16).

The study also illustrates the importance of prompt variants. Although the paper reports mainly aggregated results, the dataset includes neutral, brand explicit, and evidence seeking prompts. A source’s citation probability can change when the prompt asks for evidence or names a brand. This is a central issue for GEO research because answer systems are prompt sensitive. A measurement protocol that uses only one prompt phrasing may understate or overstate a brand’s visibility. Repeated prompts, prompt classes, and controlled prompt templates are therefore important for empirical work (17).

Another implication is that GEO measurement benefits from keeping content operations and model evaluation connected. A content team may describe a page as comprehensive, authoritative, or useful, while a model requires observable variables such as evidence density, topical coverage, structure, freshness, and independence. The modeling table creates a translation layer between those two forms of reasoning. It does not replace editorial judgment, but it makes that judgment easier to examine across many queries and source candidates. This matters because brand visibility in generated answers is not only

a matter of whether a brand has content, but whether the content is organized in a way that can be retrieved, compared, cited, and connected to the user’s information need.

7. Limitations

The analysis cannot be used to claim that any named AI answer system actually cites sources according to the reported rates. The engine labels are conditions in a simulation rather than observed products. The citation labels were generated by a controlled process that encodes assumptions about authority, evidence, structure, independence, freshness, content type, prompt variant, and query intent. As a result, the empirical results partly reflect the design of the data generation process. This is acceptable for pipeline development but not sufficient for external behavioral claims (18).

Real answer systems may use complex retrieval pipelines, proprietary indexes, browsing tools, ranking models, freshness constraints, regional settings, personalization, safety filters, and source eligibility rules. The dataset represents a fixed set of candidate sources per query. In live systems, the set of candidate sources may vary over time and may not be observable (19). Some sources may influence an answer without being cited. Some citations may be attached after answer generation or may not fully correspond to the underlying evidence used. These issues are outside the synthetic design.

The feature set is incomplete relative to real web con-

tent. It includes useful proxies such as evidence density, structured score, independence score, semantic similarity, freshness score, and domain authority, but real pages contain many additional signals. These include author reputation, factual consistency, multimedia evidence, page speed, accessibility, exact wording, entity coverage, internal linking, historical engagement, crawlability, geographic relevance, and topic specific expertise. Some of these signals are hard to observe, and some may be unavailable to outside researchers. A real GEO study would need a more detailed content extraction process and a careful audit of feature reliability (20).

The statistical tests are affected by the large sample size and the generated structure of the data. Very small differences can produce small p values when there are many observations. This is why effect sizes and practical differences are important. For example, the engine forum test has a statistically significant p value but a smaller Cramer’s V than the content type test. The interpretation should emphasize the magnitude and research relevance of differences rather than only statistical significance. In synthetic data, statistical significance also reflects the stability of the data generation process.

The modeling results may overstate what would be achievable on live data. The same process generated the features and labels, and the relationships among them may be cleaner than those in observed engine outputs. Live AI systems may vary by time, user location, account state, prompt phrasing, search index updates, and system changes. A model trained on one collection period may degrade as engine behavior changes. A realistic deployment would need repeated data collection, drift monitoring, recalibration, and evaluation on held out time periods. The current study does not address temporal generalization.

The counterfactual simulation is not causal. It changes feature values and observes model predictions, but the model is not an intervention model. In real content operations, adding a comparison table, increasing references, or reducing ad density may affect multiple features at once. It may also affect crawling, user behavior, editorial review, or eligibility for citations in ways the model does not represent. The simulation should therefore be read as a hypothesis generation tool. It can help prioritize which content features may be worth testing, but it cannot establish that a content change will cause a particular citation lift.

8. Conclusion

This paper presented a research workflow for studying generative engine optimization as a citation prediction and brand visibility problem. The study used a linked dataset of brands, queries, candidate sources, engine runs, citation events, and derived modeling features. The main unit of analysis was the query source engine prompt observation. This structure allowed the same source to be eval-

uated across query contexts, engine labels, and prompt variants. The workflow included descriptive summaries, statistical tests, supervised classifiers, ablation analysis, ranking evaluation, feature importance, and counterfactual simulations.

The results show clear differences across content formats. Research reports, comparison articles, and review pages had the highest citation rates, while product pages and pricing pages had lower citation rates (21). The content type association was stronger than the engine level differences in aggregate. Owned sources had lower citation rates than non owned sources but higher brand mention rates. This distinction suggests that GEO research should separate citation outcomes from brand mention outcomes rather than collapsing both into a single visibility concept.

The classification results show that a structured modeling table can support citation prediction under a grouped query split. Logistic regression achieved ROC-AUC 0.8890 and PR-AUC 0.8995, with accuracy 0.8079 and F1 score 0.8169. Gradient boosting and random forest also performed strongly but did not exceed the logistic regression baseline in this dataset. Ablation results showed that GEO content features and content quality features retained much of the predictive signal, while traditional SEO only, retrieval only, and authority only feature sets performed lower. These findings are consistent with the design assumption that citation readiness is related to structured, evidence oriented, and semantically aligned content signals.

The ranking and counterfactual outputs extend the analysis beyond binary classification. Ranking metrics showed that the model could place cited sources near the top of candidate lists across engine labels, with model precision at k exceeding the SERP visibility baseline in the reported comparisons. Counterfactual simulations suggested larger predicted gains for forum threads, case studies, FAQ pages, news articles, documentation, and blog posts when structure and evidence related signals were improved. These outputs are useful for forming research hypotheses and prioritizing future model evaluation.

The workflow can support future work by defining a practical schema, identifying relevant feature families, and showing how results can be reported in a transparent modeling pipeline. The same structure can be expanded with additional engines, larger query sets, richer content extraction, repeated runs, and stronger temporal validation. The central requirement is that citation outcomes, brand mentions, and candidate source features remain linked at the query source engine prompt level.

Generative engine optimization is likely to require measurement practices that differ from conventional search reporting. Citation probability, citation rank, brand mention, source independence, evidence density, and prompt sensitivity all matter when answers are synthesized rather than only listed. The work here offers one way to formalize those variables and evaluate models that predict them.

Its contribution is not a final theory of GEO behavior, but a reproducible structure for organizing data and testing models for citation based visibility outcomes.

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